Automatic Brain White Matter Hyperintensities Segmentation with Swin U-Net

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***Abstract*—This work proposes an automatic segmentation approach to detect White Matter Hyperintensities (WMH) using Fluid Attenuated Inversion Recovery (FLAIR) and the corresponding T1 weighted MRI images. In this work, we used the Swin U-Net architecture based on Transformers and compared its performance with two currently reported CNN U-Nets architectures. Sixty pairs of images and their corresponding ground truth labels were used from the MICCAI challenge. The following metrics were obtained: Dice similarity score of 0.80; lesion F1 score of 0.63; Hausdorff distance of 3.16 mm; and, 16.93 average volume difference, locating the Swin U-Net second in the ranking of the tested algorithms. However, the computational resources needed to process the imaging data were the lowest compared to the previous U-Nets. Therefore, the Swin U-Net architecture shows potential to become a promising and fast tool for segmenting medical images.**

***Keywords*—Deep Learning, Segmentation, WMH, Swin-Transformer**

I. Introduction

White Matter Hypersensitivities (WMH) are brain lesions regions which are hyperintense on T2-weighted Fluid Attenuated Inversion Recovery (FLAIR) magnetic resonance images (MRI). Small vessel disorders, strokes and depression are closely linked to WMH, and neurodegenerative diseases [1]. WMH has been extensively analysed, given its appearance in normal human ageing. A hyperintense signal is commonly referred to as brain tissue related to cerebrovascular and cardiovascular diseases [2], [3], or autoimmune and psychiatric disorders [4], [5]. WMH could also overlay other comorbidities, representing a challenge in its identification and segmentation.

Traditionally, WMH segmentation on MRI images is a manual procedure performed by expert radiologists or annotators, it is time-consuming, and since it depends on the observer, it could result in inter, and intra-expert variability [6]. MRI technologies plus Artificial Intelligence (AI) today have allowed the advancement of image segmentation tools that facilitate the relationship of WMH with various neurological diseases. WMH could be associated with motor and cognitive deficiencies [8]. The estimation is based on several properties such as volume, anatomical structures, spread, location [7], texture, tone, form and other features, which are useful to diagnose different brain pathologies. As it can be seen in Figure 1, WMH are not homogeneous and have fuzzy boundaries; hence, recognising their edges is a challenging task in the manual delineation of WMH ground truth segmentations [9]. Therefore, segmenting WMH is fundamental in exploring and understanding the aetiology, prognosis and progression of pathological diseases [10].

CNN architectures, such as the U-Net [11], [24] have been proposed to accomplish segmentation tasks automatically. In [12], WMH segmentation was performed using FLAIR and T1 with a Dice Similarity Coefficient (DSC) of 84%. Park et al. [25] also proposed a U-Net with multi-scale highlighting foregrounds, with similar results. Current approaches for automatic segmentation of WMH combine the strengths of known architectures for performing this task. The U-ResNet model combines a U-Net with a Residual network, using 2D images as input [12]; a model which can also classify between WMH and other brain lesions caused by strokes. Li et al. [15], winners of the 2017 MICCAI WMH Segmentation Challenge, have also proposed an architecture based on U-Nets with 19 fully Conv blocks in an ensemble configuration with other two U-Nets. Emerging approaches such as 3D-UNet [14], and V-Net [16] have extended their applicability in medical imaging segmentation of 3D images.

Recently, transformer-based architectures have been introduced to medical image segmentation [23]. Transformers are neural net architectures that improved the concept of self-attention [17]. Self-attention relate different positions of a single sequence in order to generate a representation of the same sequence. Each layer of self-attention is composed by Queries, Keys and Values. These concepts are derived from retrieval systems. As an analogy, in a Youtube search, the search engine use the inserted query to map it against a group of keys, like titles or descriptions, that are associated to a group of candidates videos in a database. Values are the best fit result for that search. Transformers were initially used in natural language processing, however, today they are also utilised in the context of computer vision [18]. Vision transformers divide images in patches, hence they can be input as sequences in a neural network. For medical images analysis, the use of transformer architectures such as the Swin U-Nets in the context of organ segmentation [23], or Trans U-Nets for cardiac diagnosis [19] have shown promising results. This proposal aims at improving the segmentation performance of WMH using a recent vision-transformer based architecture, as compared to the predefined metrics established by the MICCAI WMH Segmentation Challenge. Additionally, this proposal aims to improve the segmentation training and inference times.

II. Methodology

A. The dataset

For this work, we have used the public dataset from the MICCAI WMH Challenge [20]. This dataset comprises 60 image pairs from three institutions, consisting of FLAIR and T1 images and WMH segmented masks as the ground truth. The process of image segmentation to obtain the masks was performed by two expert observers, O_1 and O_2, following the **ST**andards for **R**eport**I**ng **V**ascular changes on n**E**uroimaging (**STRIVE**) conventions [21]. Each WMH lesion was contoured by the observed O_1 while a peer review over the manual delineations was conducted by the second observer O_2.

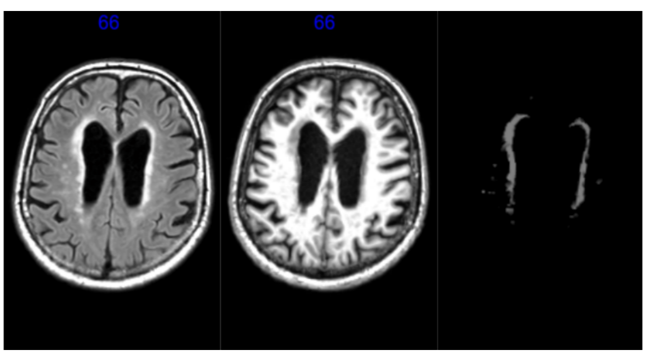


Fig. 1. An instance of the dataset is composed by a FLAIR and a T1-weighted image, and a manually segmented mask of the WMH.

More details about the dataset are illustrated in Table I

B. Data Preparation

Images and masks from the native space were pre-processed using the python ANTs library [29] and the scipy library [30], through the following steps: A bias field correction to correct inhomogeneities produced by the magnetic field; a denoising step to remove image noise while preserving

TABLE I
Detailed Description of WMH Dataset

Institution	Scanner	# images
UMC Utrecht	3 T Philips Achieva	20
NUHS Singapore	3 T Siemens TrioTim	20
VU Amsterdam	3 T GE Signa HDxt	20

the integrity of the remaining image information; an image smoothing with Gaussian kernel to smooth images and reduce the effects of small clusters; then an intensity normalization to reduce variability between the different scanner technologies. A masking step was necessary to fill any holes in the images; and, the intensity normalisation was performed by subtracting the mean and dividing them by the standard deviation.

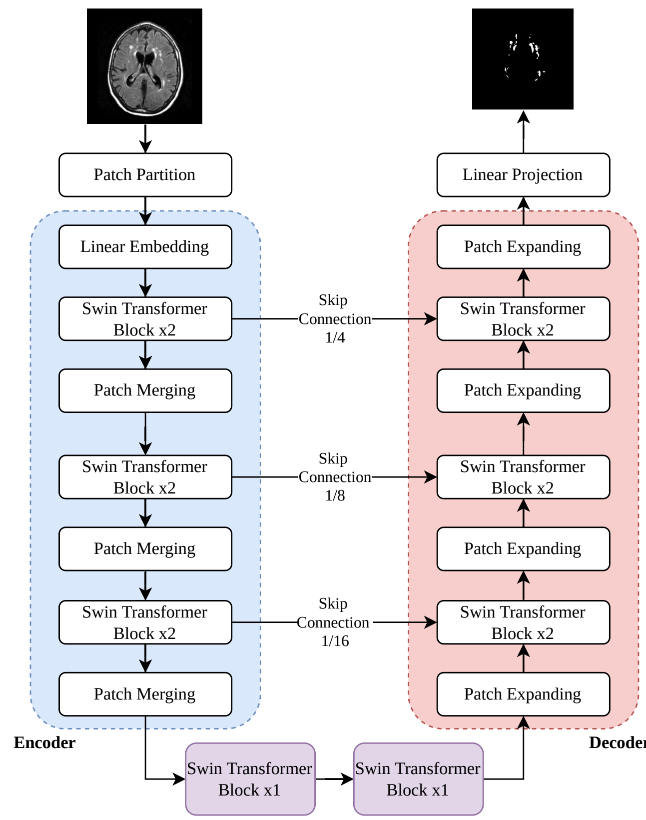


Fig. 2. The proposed Swin U-Net architecture

Next, for each 3D volume (FLAIR and T1 images and ground truth mask) of each subject, 48 - 2D axial images were generated, which were resized to 224x224. Then, a data augmentation strategy was implemented to increase the imaging data. Random flipping, angle rotation and zoom were applied to the training images to increase, not only the dataset size, but also the robustness of the model.

C. Network Architecture

In this approach, a Swin U-Net [23] architecture has been used, as it is illustrated in Figure 2. The model comprises four main sections: An encoder, a bottleneck, a decoder and a set of skip connections. The encoder section is responsible for splitting the FLAIR image of size $W \times H$ into non-overlapping patches of 7×7. Then, a downsampling procedure was applied to allow the feature extraction process to be performed by the Swin Transformer Block with the Patch Merging layer. The bottleneck section corresponds to two Swin Transformer blocks of the exact resolution and feature dimension. The decoder section portrays a symmetrical structure with the encoder. Then, the upsampling step was performed by a set of Patch Expanding layers and the Swin Transformer Blocks. Three skip connections were added in this process to ameliorate the loss of information produced by downsampling. The architecture ends with a linear projection layer, which produces a tensor of size $W \times H \times C$, where C is the number of segments to be generated.

D. Implementation details

The proposed model was built in Python 3.7 with the PyTorch 1.x framework. For testing purposes, we have fragmented the dataset using an 85 - 15 % split, where 51 images were used in training and 9 for testing.

As for the training and testing procedures, we used a computer with a 6th Gen Intel processor, 16 GB of RAM and a Nvidia 1060 GPU with 6GB of VRAM, running Ubuntu 18.04. In the training step, the model parameters were initialized using the pre-trained weights from an organ segmenter and updated using cross-entropy and Dice loss [22]. Regarding the optimizer settings, the Adam Stochastic Gradient Descent [13] was used with a learning rate set to 0.001 and a momentum to 0.9. A batch size of 24 was used in 100 epochs.

E. Evaluation metrics

To evaluate the performance of our model, we have considered five metrics, as implemented in [24] for the WMH Segmentation challenge. Those metrics are: Dice Similarity Coefficient (DSC), Hausdorff Distance (HD), Average Volume Difference (AVD), F1 score and Recall. More extensive details about the metrics can be found in [26].

Dice similarity coefficient

The Dice Similarity Coefficient (DSC) was used to measure the overlap between a given ground-truth segmentation map *G* and another segmentation map *P*, which is generated by the neural network model, as expressed in Eq. (1).

$$DSC = 2\frac{|G \cup P|}{|G| + |P|} \tag{1}$$

Hausdorff Distance.

Hausdorff Distance [32] measures how far two subsets of a metric space are from each other. More informally it measures the greatest of all distances from a point in one set to the closest poit in the other set. This metric is calculated as follows:

$$d_H(\mathcal{X}, \mathcal{Y}) = \max\{\sup_{x \in \mathcal{X}} d(x, \mathcal{Y}), \sup_{y \in \mathcal{Y}} d(y, \mathcal{X})\}, \tag{2}$$

where $\mathcal{X}, \mathcal{Y}$ are non-empty sets of metric space (M, d).

Average Volume Difference

The Average Volume Difference (AVD) measures the percentage difference in the volume of the manual segmentation lesions against model segmentation. More formally, given V_g and V_p the respective volumes of *G* and *P*, the Average Volume Difference is defined as:

$$AVD = \frac{|V_g - V_p|}{V_g}. \tag{3}$$

Recall

Recall measures the ability of a model to find all relevant cases within the data. More formally, let N_g be the number of individual lesions delineated by the ground truth *G*, and N_p the correct number of detected lesions comparing both *G* and *P*. Therefore, this metric can be expressed as in Eq. (4):

$$Recall = \frac{N_p}{N_g}. \tag{4}$$

F1 Score

The F1 score is a way to combine Recall with precision and is defined as the harmonic mean of both. More formally, given N_p as the correct number of detected lesions comparing both *G* and *P*. Let N_f be the wrong lesions in *P*. Therefore, the F1 score can be defined as:

$$F1 = \frac{N_p}{N_p + N_f}. \tag{5}$$

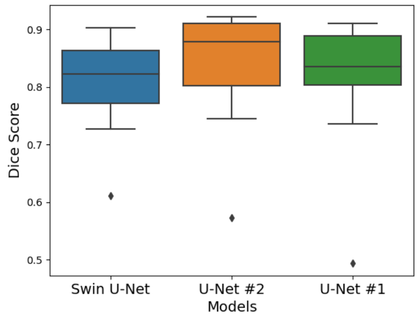
III. Results and Discussion

A. Evaluation metrics results

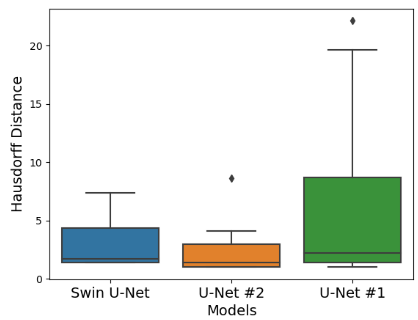
For evaluating the performance of the proposed Swin U-Net, the following two CNN U-Nets have also been implemented and their corresponding scores compared. The model identified as U-Net #1 as described in [26] and ilustrated in Figure 4, was designed to handle two input channels for the FLAIR and T1 images. This network was built using 21 convolutional layers, 3 upsampling layers, and 3 pooling layers. For the first two layers we used 5x5 convolutional filters, while the rest employed 3x3 convolutional filters. A Rectified Linear Unit (RELU) [31] activation function was also applied after each convolutional layer.

The second model has been identified as U-Net #2, which was proposed in a previous work [24], with an architecture similar to the model shown in Figure 4, but with a different input; this model processes FLAIR images only as inputs. Aditionally, the convolutional layers with 96 filters are discarded in the contractive path as well as the expansive path.

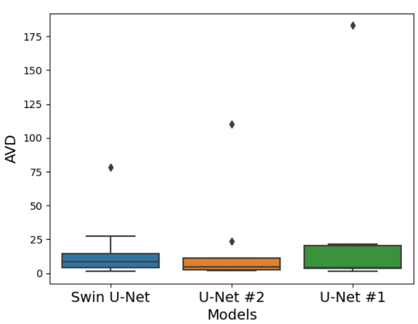
Table II shows the evaluation results performed on test data. As it can be seen in the table, the Swin U-Net proposed



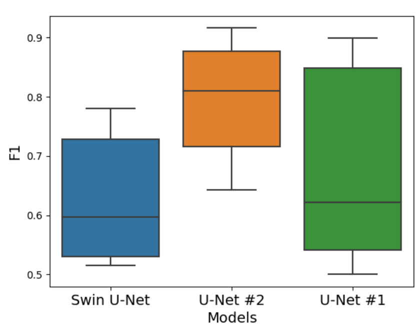
(a) Dice score coefficient



(b) Hausdorff distance



(c) Average volume difference



(d) F1

Fig. 3. Box-plot comparison of the evaluation metrics for the implemented segmentation architectures.

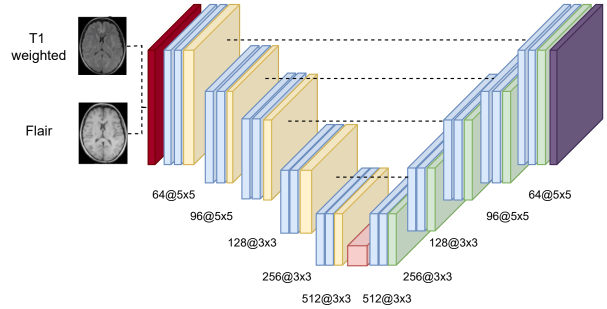


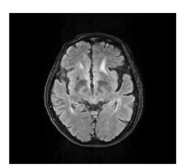
Fig. 4. Neural network architecture proposed for U-Net #1

reached better performance scores on the AVD compared to the other U-Nets architectures. As for the Dice Score and the Hausdorff distance metrics, the Swin U-Net scored similarly to the CNN U-Nets. However, comparing the Recall and F1 scores of our approach, it was lower than the other CNN U-Net based reported scores, as it can be observed in Table II.

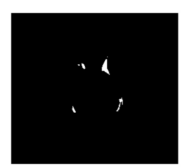
Figure 3 shows a comparison of the performance achieved by each architecture based on each of the selected metrics. As observed in this figure, the U-Net #2 architecture performed better than the rest of models in terms of the Dice Score coefficient. A similar behaviour can be observed comparing the Hausdorff distance and Average Volume difference. However, the Swin U-Net remained competitive against U-Net #2 in the three metrics. In F1, U-Net #2 clearly outperformed the other two models.

TABLE II
MEAN SEGMENTATION METRICS OF DIFFERENT APPROACHES IN TEST DATA ON THE MICCAI-2017 WMH DATASET

Architecture	**DSC**	**HD**	**AVD**	**Recall**	**F1**
Swin U-Net	0.80	3.16	16.93	0.72	0.63
U-Net #1 [26]	0.80	6.76	28.09	0.88	0.68
U-Net #2 [24]	0.83	2.54	18.47	0.81	0.80



(a) FLAIR image



(b) Manual WMH segmentation

(c) Output of the Swin UNet model

Fig. 5. An instance comparison between the ground-truth mask and the automatic segmentation from the proposed model.

B. Inference and training times

Table III shows the results of both, the training and inference times of the three models. All models were trained twice to estimate the training time, then a mean time was calculated. As for estimating the inference time, we used 10 images and followed the same procedure.

As observed, training time in the Swin U-Net model was much lower, with a mean time of two hours as compared to the other CNN U-Nets, which took around 8 hours. However, during inference, a similar outcome was reached. The inference time of the transformers method was about 2.4 seconds, as compared to 11.6 and 17.7 seconds of U-Net #1 and U-Net #2, respectively.

Reducing the training time to obtain a faster model convergence is one of the biggest challenges in Deep Learning (DL) models. Hence, a solution with noticeable performance has been proposed, showing comparable results as with other approaches, such as applying pooling layers to reduce the parameters dimensionality [27], or combining them with additional strides [28]. DL approaches with low training times can also be reached by excessive regularisation or a sudden increase of the model learning rate. However, this proposed approach shows promising results, not only with a competitive performance, but with a significant lower training time.

TABLE III
INFERENCE AND TRAINING TIMES

Architecture	Training time	Inference time
Swin U-Net	103 minutes	2.38 seconds
U-Net #1 [26]	490 minutes	11.60 seconds
U-Net #2 [24]	523 minutes	17.72 seconds

IV. CONCLUSIONS

We have developed a fine-tuned Swin U-Net, a transformer-based neural network model architecture, to automatically segment FLAIR images with White Matter Hyperintensities. The resulting model showed competitive results in three out of the five metrics chosen to evaluate its performance, as compared to other two CNN U-Nets based models.

The transformer-based model has shorter inference and training times than its convolutional U-Nets counterparts. Therefore, the Swin U-Net shows promising performance in segmenting MRI, which could be implemented in medical imaging segmentation algorithms, requiring lower computational resources.

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